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Social media use among Italian older adults: a weighted analysis of Facebook survey data

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ABSTRACT

Older adults are increasingly active online, yet their social media engagement remains understudied. This research explores Facebook use among Italians aged 60 and over, examining motivations, habits, and digital literacy factors while also assessing the effectiveness of Meta advertisements for survey recruitment. Based on 652 responses collected through quota sampling, the study applies weighting techniques to adjust for sample bias, with raking proving the most effective.

The use of Facebook primarily for communication with friends and family among older adults varies by age, digital habits, and self-perception. Individuals aged 60–74 are less likely to use it for this purpose than those 75+. Self-learners tend to rely more on Facebook for social communication. Additionally, those who do not verify information offline are more dependent on Facebook for communication. These findings provide insights for digital engagement strategies and surveys targeting older age groups in Italy.

INTRODUCTION

This study investigates the motivations, habits, and digital literacy factors shaping Facebook use among Italians aged 60 and over. Given the increasing reliance on digital communication, it explores whether Facebook can serve as a platform for fostering social engagement and reducing loneliness in this demographic (Musaio, 2022). Additionally, the research examines the feasibility of Meta advertisements for recruiting older survey participants, addressing a key methodological challenge in studying this age group.

PROBLEM

Older adults represent an increasingly active demographic in the digital world, yet their engagement with social media remains underexplored. While most studies focus on younger users, understanding how older adults interact with online platforms is crucial for developing policies that promote digital inclusion and social well-being. This study aims to fill this gap by analyzing whether Facebook primarily functions as a communication tool for reducing social isolation or as a space for other types of engagement. It also evaluates the effectiveness of digital recruitment strategies, offering insights into how online surveys can better reach this population.

DATA

Our work is based on a survey conducted through a questionnaire distributed via Facebook's Meta platform, which enabled direct contact with the over-60 population. The survey was advertised between 13th and 20th August 2024, collecting a total of 652 valid responses with a budget of 600 Euros, following a quota sampling.

The questionnaire was structured into six sections, covering sociodemographics (e.g., gender, location, marital status, employment), internet use (devices, frequency, social media

platforms, motivations), Facebook-specific behavior (usage patterns, confidence in use), online administrative procedures (preferences for online vs. offline tasks), online security awareness (fake news, information verification, cookies), and social media self-perception (respondents selected an image representing their relationship with social media).

DATA CLEANING & VALIDATION

Although our approach ensured wide dissemination and rapid data collection at low cost, it introduced a partial distortion in the sample. The questionnaire reached elderly individuals who were already active Internet users and relatively familiar with digital technologies. As a result, the sample does not fully reflect the reality of the Italian elderly population, nor that of Italian seniors who use the Internet, as not all older individuals with Internet access possess strong digital skills or have a Facebook account.

To make the data representative of the Italian 60+ population active on the Internet, we used data from the Multipurpose Survey on Households: Aspects of Daily Life (AVQ – *Aspetti della Vita Quotidiana*), an annual, nationally representative survey conducted by the Italian National Institute of Statistics (ISTAT, 2023).

Moreover, since the dataset was collected firsthand, extensive data cleaning was required to ensure its accuracy and reliability. The process included imputing missing values where applicable, recoding categorical variables for standardization, and identifying inconsistencies and potential biases in survey responses.

ANALYSIS

METHODS

Statistical analysis was conducted using **SAS® OnDemand for Academics**, with all tests performed at a 5% significance level.

One of the most challenging aspects of the process was addressing a significant bias in the representation of the elderly. In our sample, most older participants were active Internet users, whereas in the general Italian population this is far from the norm. Therefore, as a first step, a bias correction was applied to adjust the sample and better reflect the Italian population of Internet users aged 60 and over. A crucial initial step was identifying the benchmark variables present in both our dataset and the AVQ, evaluating which weighting strategy would produce the most accurate results. This discrepancy required a sophisticated approach, leading us to recur to a weighting strategy to produce more accurate results. The initial approach, based solely on socioeconomic and demographic variables, failed to sufficiently correct the sample distortion. Thus, we integrated variables related to digital skills into the post-stratification process. The inclusion of these variables was a fundamental breakthrough, as it ultimately eliminated the distortion and ensured that, after applying the weights, the AVQ benchmark variables closely matched those in our dataset. Correcting the sample distortion was a complex and iterative process in which we tested three weighting methods: post-stratification (**PROC FREQ**), raking (iterative proportional fitting) (**PROC RANK**), and quasi-randomization (Valliant, 2020) (**PROC SURVEY SELECT**). All subsequent analyses were based on the weighted dataset, ensuring that our findings accurately represent the target population.

The second phase of the analysis focused on exploring relationships between key variables. Bivariate association tests and chi-square tests (**PROC FREQ** with the **WEIGHT** option) were employed to assess statistical significance and identify preliminary significant associations among variables.

In the final phase, a logistic regression model (**PROC LOGISTIC** with the **WEIGHT** option) with stepwise selection to investigate the role of Facebook in addressing social isolation among older adults. The model specifically aimed to identify factors influencing whether seniors use

Facebook primarily for communication with friends and family or for other purposes, aligning with our research focus on social engagement and digital inclusion.

RESULTS

Effectiveness of Weighting Methods

Although all three weighting methods provided reasonable results, raking implemented with **PROC RANK**, emerged as the most effective approach. This outcome underscores a key insight from our work: the success of a weighting strategy is not solely determined by the method used but rather by the careful selection of post-stratification variables. The process was far from straightforward, requiring continuous refinements and in-depth methodological considerations, highlighting the intricate nature of sample adjustment in the presence of nontraditional biases.

Social Media Use Among Older Adults

As shown in Figure 1, the weighted percentages indicate that among Italian elderly internet users, excluding Facebook, WhatsApp is the most widely used platform (88%), followed by Instagram (34%). Pinterest (10%) and Telegram (8%) have lower adoption, while Twitter and TikTok usage remains below 5%. These findings highlight the dominant role of WhatsApp as a communication tool among older adults, whereas other social media platforms show significantly lower engagement in this demographic. Additionally, they confirm that this age group primarily relies on a few select platforms for online interaction.

Facebook Usage Patterns

Older adults primarily use Facebook to communicate with friends and family, as indicated by the highest percentage of "Frequent Use" responses (40.2%), while activities such as playing games, making new acquaintances, live streaming, and using the marketplace are significantly less common, as shown by the high percentage of "Never" responses. These findings suggest that, for this demographic, Facebook serves primarily as a tool for social connection and information consumption rather than for commercial activities or content creation.

Multivariate analysis

To further investigate the factors influencing Facebook usage among older adults, a logistic regression model was applied using **PROC LOGISTIC** in SAS. The dependent variable distinguishes between individuals who use Facebook primarily for communication with friends and family and those who use it for other purposes. A stepwise selection approach was employed to identify the most relevant predictors, using an entry significance level (**SLENTRY**) and a stay significance level (**SLSTAY**) of 0.05. The regression model incorporates a broad set of predictors, including demographic characteristics (age, gender, marital status), digital skills, online behavior, exposure to digital advertisements, and platform engagement patterns. All categorical variables were declared using the **CLASS statement** with reference coding (**param=ref**). The model was weighted to ensure representativeness.

The logistic regression analysis provides key insights into the determinants of Facebook usage for communication among older adults.

Age plays a significant role, as individuals aged 60–64 and 65–74 are significantly less likely to use Facebook for communication compared to those aged 75 and older. This suggests that younger elderly users could tend to diversify their digital communication tools, such as WhatsApp and email.

Learning methods also influence usage patterns. Those who did not learn Facebook from friends are more likely to use it for social communication, indicating that self-learners may rely more on Facebook for maintaining connections.

The timing of Facebook download appears to be another key factor. Users who downloaded the platform more than four years ago are less likely to use it for communication, possibly reflecting shifts in platform functionality and evolving user habits.

In our campaign, participants were divided into quotas, with each group shown a specific advertisement image based on their assigned category. The images depicted different scenarios: intergenerational interactions (Org), technology-related fear (Fear), older users with a neutral relationship with social media (Old), a family setting with grandparents and grandchildren (Grand), and a neutral representation of social network use (Neutr). The results indicate that participants exposed to images of intergenerational connections, older users, or a neutral portrayal of social networks were less likely to use Facebook for communication.

Self-perception in terms of digital identity also plays a role in online behavior. Participants were asked to describe their relationship with social media by selecting the image that best represented them among four available options. Each image symbolized a different digital identity: an organized approach to social media (Org), a perception of social media as overwhelming or anxiety-inducing (Fear), older users with a neutral relationship with social media (Old), and a neutral stance towards digital interactions (Neutr). The findings reveal that respondents who identified as 'organized' or 'neutral' digital users were more likely to use Facebook for communication.

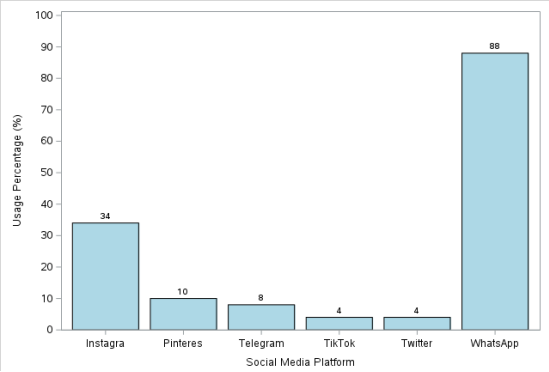

Figure 1. Bar chart of the weighted univariate percentage distribution of social media usage among Italian older adults

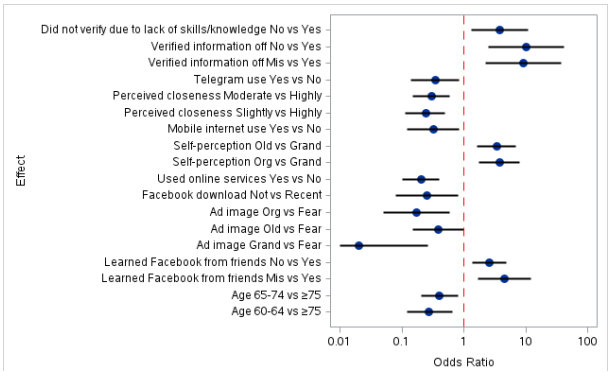

Figure 2. Statistically significant Odds Ratios and confidence intervals for the logit model fitted with PROC LOGISTIC. Response variable: Use of Facebook for communication with friends and family.

Different aspects of online behavior further contribute to these patterns. Individuals who use online administrative services are less likely to rely on Facebook for communication, suggesting that digitally skilled users may prefer other online tools. Similarly, those who have recently started using mobile internet are less likely to use Facebook for communication, possibly because they have not yet integrated the platform into their digital habits. Additionally, Telegram users exhibit a lower likelihood of using Facebook for communication, indicating a potential substitution effect between messaging apps.

Information verification habits also play a role. Users who do not verify information offline are significantly more likely to rely on Facebook for communication. This suggests that lower engagement in critically evaluating online content may lead to greater dependence on Facebook for social interactions.

For a detailed breakdown of parameter estimates and odds ratios, refer to Tables A.1 and A.2 in the Appendix. Statistically significant odds ratios and confidence intervals are presented in Figure 2.

SUGGESTIONS FOR FUTURE STUDIES

Future research could adopt a longitudinal perspective to track changes in social media behavior, compare findings across different cultural contexts, and assess the impact of digital literacy programs on online engagement. Additionally, exploring more inclusive recruitment strategies beyond social media ads would also help reach a more diverse segment of the aging population.

CONCLUSION

This study highlights the complexity of social media engagement among older adults, showing that Facebook remains a key communication tool, with its role varying based on digital habits and self-perception. While online surveys effectively reach digitally active seniors, they may not fully represent the entire older population. The findings emphasize the importance of weighting techniques in reducing selection bias, allowing for a better understanding of the profiles of those who primarily use Facebook for communication. These insights provide a solid foundation for future research on aging and technology.

REFERENCES

Ali, S.H., Foreman, J., Capasso, A., Jones, A.M., Tozan, Y., and R.J. DiClemente. 2020. "Social Media as a Recruitment Platform for a Nationwide Online Survey of COVID-19 Knowledge, Beliefs, and Practices in the United States: Methodology and Feasibility Analysis." *BMC Medical Research Methodology,* 20:116.

ISTAT. 2023. "Microdati - Multiscopo sulle Famiglie: Aspetti della Vita Quotidiana." *Italian National Institute of Statistics.* Available at https://www.istat.it/microdati/multiscopo-sulle-famiglie-aspetti-della-vita-quotidiana/.

Izrael, D., Hoaglin, D.C., and M.P. Battaglia. 2000. "A SAS Macro for Balancing a Weighted Sample." *Proceedings of the Twenty-Fifth Annual SAS Users Group International Conference.* Cary, NC: SAS Institute.

Musaio, M. (2022). Dentro la solitudine degli anziani: da una narrazione in ambito geriatrico ad una rilettura pedagogica. *Journal of Health Care Education in Practice*, 1, 25-40.

SAS Institute Inc. 2017. "To Rake or Not to Rake Is Not the Question Anymore with the Enhanced Raking Macro." *Proceedings of the SAS Global Forum 2017 Conference.* Cary, NC: SAS Institute. Available at https://support.sas.com/resources/papers/proceedings17/0470-2017-poster.pdf.

SAS Institute Inc. 2017. "The LOGISTIC Procedure." *SAS/STAT® 14.1 User's Guide*. Cary, NC: SAS Institute Inc.

APPENDIX

Coefficient of	Value	Estimate	SE	Chi-quadrato di Wald	p-value
Intercept		**3.29**	1.31	6.14	0.01
Age	60–64 years	**-1.31**	0.44	8.96	<.01
	65–74 years	**-0.93**	0.35	6.95	0.01
Learned Facebook from friends	Missing	**1.51**	0.51	8.92	<.01
	No	**0.93**	0.33	8.21	<.01
Ad image	Neutr	1.42	1.13	1.61	0.21
	Nipo	**-3.71**	1.20	9.57	<.01
	Old	**-0.97**	0.49	3.94	0.05
	Org	**-1.76**	0.62	8.04	<.01
Facebook download	Not Recent	**-1.40**	0.60	5.50	0.02
Used online services (12 months)	Yes	**-1.61**	0.35	21.25	<.01
Social media self-perception	Org	**1.32**	0.39	11.61	<.01
	Fear	-0.75	0.53	2.03	0.15
	Old	**1.21**	0.38	10.34	<.01
Mobile internet use (3 months)	Yes	**-1.14**	0.49	5.36	0.02
Perceived closeness via mobile	Slightly	**-1.45**	0.37	15.06	<.01
	Moderately	**-1.21**	0.34	12.73	<.01
Telegram use (last month)	Yes	**-1.08**	0.46	5.52	0.02
Verified information offline	Missing	**2.22**	0.72	9.50	<.01
	No	**2.31**	0.72	10.33	<.01
Did not verify due to lack of skills/knowledge	Missing	0.57	0.50	1.33	0.25
	No	**1.32**	0.54	6.02	<.01

Table A.1. Parameter estimates for the logit model fitted with PROC LOGISTIC. Response Variable: *Use of Facebook primarily for communication with friends and family (vs. other purposes).*In bold: statistically significant estimates at the 5% level.

Coefficient of	Effect	Odds ratios	95% Wald Confidence Limits
Age	60–64 years compared to ≥75	**0.27**	0.12 - 0.64
	65–74 years compared to ≥75	**0.40**	0.20 - 0.79
Learned Facebook from friends	Missing compared to Yes	**4.54**	1.68 - 12.23
	No compared to Yes	**2.54**	1.34 - 4.81
Ad image	Neutr compared to Fear	4.17	0.46 - 37.98
	Grand compared to Fear	**0.02**	0.01 - 0.26
	Old compared to Fear	**0.38**	0.15 - 0.99
	Org compared to Fear	**0.17**	0.05 - 0.58
Facebook download	Not Recent compared to Recent	**0.25**	0.08 - 0.79
Used online services (12 months)	Yes compared to No	**0.20**	0.10 - 0.40
Social media self-perception	Org compared to Grand	**3.76**	1.76 - 8.05
	Fear compared to Grand	0.47	0.17 - 1.33
	Old compared to Grand	**3.35**	1.60 - 7.00
Mobile internet use (3 months)	Yes compared to No	**0.32**	0.12 - 0.84
Perceived closeness via mobile	Slightly compared to Highly	**0.24**	0.11 - 0.49
	Moderately compared to Highly	**0.30**	0.15 - 0.58
Telegram use (last month)	Yes compared to No	**0.34**	0.14 - 0.84
Verified information offline	Missing compared to Yes	**9.21**	2.24 - 37.81
	No compared to Yes	**10.07**	2.46 - 41.21
Did not verify due to lack of skills/knowledge	Missing compared to Yes	1.77	0.67 - 4.69
	No compared to Yes	**3.76**	1.31 - 10.80

Table A.2. Odds Ratios and Confidence Intervals for the Logit Model Fitted with PROC LOGISTIC. Response Variable: *Use of Facebook primarily for communication with friends and family (vs. other purposes).* In Bold: Statistically Significant Estimates at the 5% Level.